

Trainings ▾

Preparatory Steps

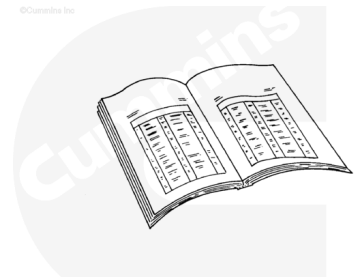
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

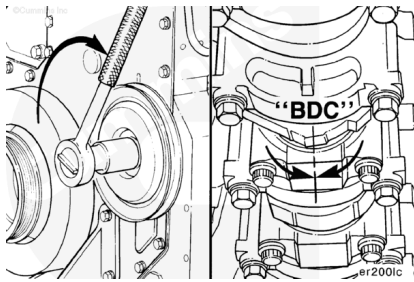
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-046-tr.html>)



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Remove

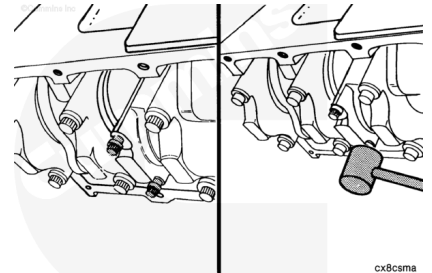
Use the accessory drive pulley to rotate the crankshaft so the connecting rod to be removed is at bottom dead center.



Loosen the connecting rod capscrews.

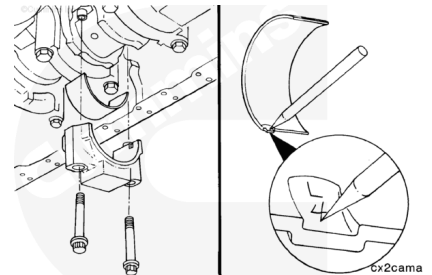
Do **not** remove the capscrews.

Hit the connecting rod capscrews with a rubber hammer to loosen the rod caps from the dowel rings.



Remove the capscrews and rod caps.

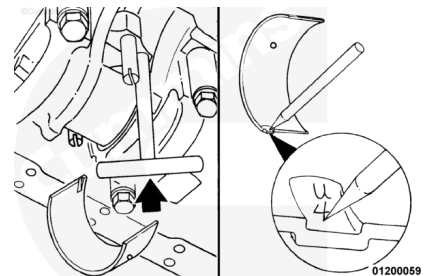
Remove the lower bearing shell from the rod cap. Mark it with the letter "L" (for lower) and the cylinder number it was removed from.



Install two connecting rod guide pins, Part Number 3376038.

Use a t-handle piston pusher to push the rod away from the crankshaft to allow the upper bearing shell to be removed.

Remove the upper bearing shell. Mark it with the letter "U" (upper) and the cylinder number it was removed from.



Clean and Inspect for Reuse

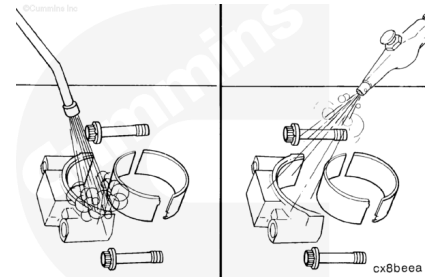
Center Sump Pan

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ WARNING ⚠

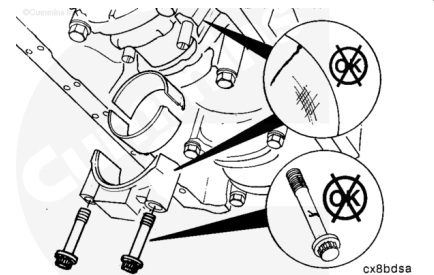
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Steam clean the parts.

Dry with compressed air.

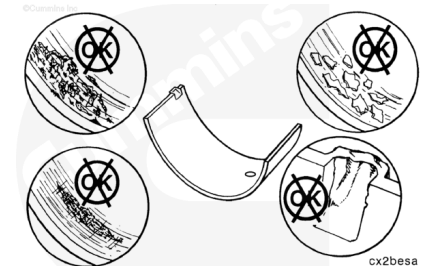
Inspect the rod caps, connecting rod bearing saddles and capscrews for nicks, cracks, burrs, scratches or frets.



Inspect the bearings for damage.

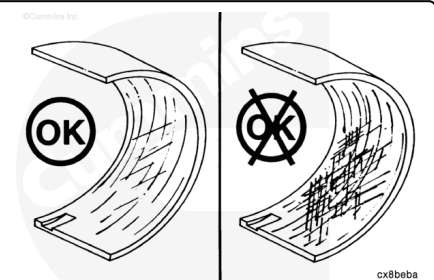
Replace any bearings with the following damage:

- Pits
- Flakes
- Corrosion
- Lock tang damage
- Scratches (deep enough to be felt with a fingernail).



Normal bearing wear produces a smooth finish that will wear into the copper lining. Exposed copper does **not** always indicate worn bearings. Refer to the Parts Reuse Guidelines, Bulletin 3810303.

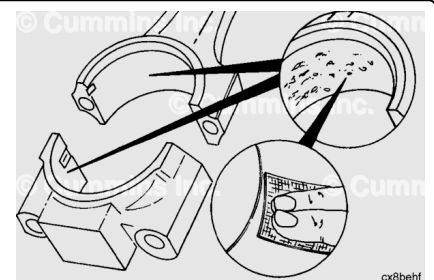
If large areas of copper lining are visible in the bearings before the engine has accumulated 241,000 km [150,000 mi] or 3750 hours, inspect the engine for contamination from fine dirt particles and correct the problem.



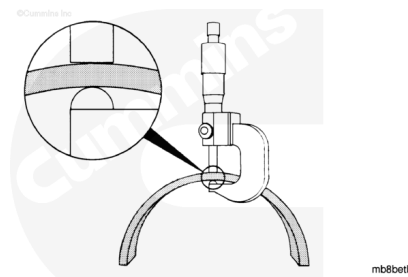
Inspect the bearing shell seating surface for nicks or burrs.

If nicks or burrs can **not** be removed with Scotch-Brite® 7448 or equivalent, the bearings **must** be replaced.

For more detailed information on bearing damage, refer to the Analysis and Prevention of Bearing Failures Manual, Bulletin 3810387.



Use an outside diameter ball-tipped micrometer to measure the rod bearing thickness.



Standard Connecting Rod Bearing Thickness

mm		in
2.430	MIN	0.0957
2.473	MAX	0.0974

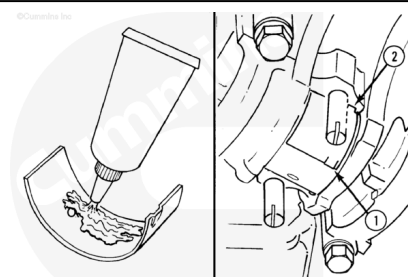
Install

Used bearings **must** be installed in the same location from which they were removed.

Use Lubriplate™ 105, or equivalent, to coat the inside diameter of the bearing shell.

Install the upper bearing shell in the connecting rod with the tang (1) of the bearing in the slot (2) of the rod.

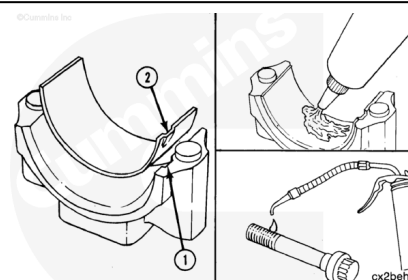
Remove the two connecting rod guide pins, Part Number 3376038.



Install the bearing shell in the connecting rod cap with the tang (2) of the bearing in the slot (1) of the cap.

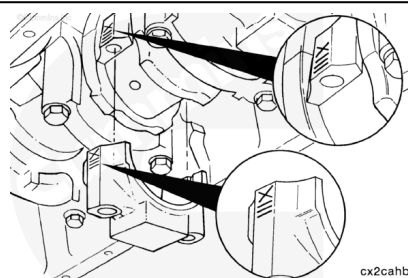
Use Lubriplate™ 105, or equivalent, to coat the inside diameter of the bearing shell.

Use clean 15W-40 oil to lubricate under the head and the threads of the connecting rod capscrews.



The connecting rod and cap **must** be matched with the same number. The tang **must** be toward the camshaft side of the engine.

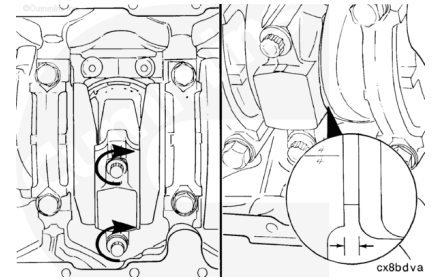
Install the connecting rod caps and capscrews.



Tighten the capscrews in alternating sequence to the following torque values:

Torque Value:

1. 68 n•m [50 ft-lb]
2. 142 n•m [105 ft-lb]
3. 210 n•m [155 ft-lb]
4. Loosen completely and repeat steps 1 through 3

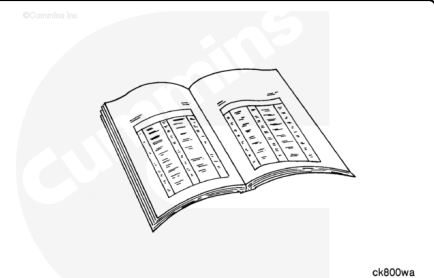


Measure the connecting rod side clearance. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html)

Finishing Steps

- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-046-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)
- Fill the lubricating oil pan. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html)
- Operate the engine to normal operating temperature and check for leaks.



Note : The engine must have adequate oil pressure within 15 seconds after starting. If the warning light indicating low oil pressure has not gone out or there is no oil pressure indicated on a gauge within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.